

PROJECT REPORT: US TRAFFIC ACCIDENT ANALYSIS

ProDigy Infotech Task-05 | Data Science Internship

1. Executive Summary

This project analyzes a large-scale dataset of traffic accidents in the United States to identify key patterns related to **road conditions**, **weather**, and **time of day**. By leveraging Python's data science stack, we processed millions of records to identify accident hotspots and primary contributing factors to road incidents.

2. Project Pipeline

The project followed a structured data science workflow to ensure high-quality insights:

- **Environment Setup:** Loading the US-Accidents dataset using Pandas with memory optimization techniques.
- **Data Cleaning:** Handling missing values in environmental columns and removing redundant fields like End_Lat and End_Lng.
- **Feature Engineering:** Extracting Hour, Day of Week, and Month from timestamps to enable temporal analysis.
- **Exploratory Data Analysis (EDA):** Performing univariate and bivariate analysis to find correlations between accidents and external factors.
- **Visualization:** Using Seaborn and Matplotlib to create heatmaps and distribution charts for storytelling.

3. Key Insights & Findings

A. Temporal Patterns (The "When")

The Rush Hour Effect: Weekday data reveals a distinct "Dual-Peak" pattern. Accidents spike significantly between 7 AM – 9 AM and 4 PM – 6 PM, coinciding with professional commute hours. Conversely, weekends show a single broad peak in the afternoon.

B. Environmental Factors (The "Weather")

The Volume Paradox: Over 70% of accidents occur during "Fair" or "Clear" conditions. This suggests that high traffic volume and driver behavior on good roads are more frequent causes than severe weather itself. However, accidents during "Rain" or "Fog" often show higher severity scores.

C. Infrastructure Impact (The "Where")

High-Conflict Zones: Traffic Signals, Junctions, and Crossings are the top three infrastructure features associated with accidents. Geographically, California, Florida, and Texas lead the nation in accident frequency due to population density and highway network scale.

4. Technical Implementation

Stage	Activity	Tools Used
Data Handling	Memory-optimized CSV loading and cleaning	Pandas, NumPy
Feature Extraction	Time-series parsing and column derivation	DateTime Library
Visualization	Heatmaps, Pie Charts, and Bar Plots	Seaborn, Matplotlib

5. Conclusion & Recommendations

- **Dynamic Deployment:** Traffic authorities should increase patrol presence during weekday peak commuting hours (8 AM and 5 PM).
- **Infrastructure Focus:** Safety audits should be prioritized at high-volume junctions in CA, FL, and TX to optimize merge lane safety.
- **Behavioral Campaigns:** Since most accidents occur in clear weather, public awareness should focus on distracted driving rather than just weather-related hazards.